Oscar A. Flores-Gaitán

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Research Interests

- Precision Doppler detection and characterization of exoplanetary systems.
- Modeling and mitigating stellar activity in radial velocity and spectroscopic time series.
- Developing and applying machine learning and statistical techniques to astronomical data.

Education

2026 – Present **Ph.D. in Astronomy & Astrophysics**, *Department of Physics & Astronomy, University of Delaware, Newark, DE*

2022 – 2026 **Licentiate in Physics**, *Department of Physics, Universidad del Valle de Guatemala, Guatemala City*

Honors: *Magna Cum Laude*

- Thesis: "*Decoding the Barnard's Star Planetary System: A Time Series Based Analysis of the Planetary Architecture*"

Publications

First-Author

2027 **Flores-Gaitán, O.A.**, Dodson-Robinson, S., *"Detecting Harmonic Coupling in Stellar Activity using Dual-Frequency Autocoherence"*, [Manuscript in preparation, 2027].

2026 **Flores-Gaitán, O.A.**, Dodson-Robinson, S., Ramirez Delgado, V., *"A Dynamically Stable Three-Planet Architecture for Barnard's Star"*, [To be submitted, 2026].

Second- & Third-Author

2027 Cambronero Jiménez, I., Dodson-Robinson, S., **Flores-Gaitán, O.A.**, *"Cepstrum Analysis of Stellar Activity"*, [Manuscript in preparation, 2027].

Co-Authored & Contributor

2026 Zhao, L.L. et al., **including Flores-Gaitán, O.A.**, *"The Extreme Stellar-Signals Project IV"*, [Manuscript in preparation, 2026].

2026 Dressing, C.D., Adams, D., Alecian, E. et al., **including Flores-Gaitán, O.A.**, *"Early Exploration of the Scientific Discovery Space for the Habitable Worlds Observatory"*, arXiv preprint (2026). arXiv:2608.11294.

Theses

2026 **Flores-Gaitán, O.A.**, *"Decoding the Barnard's Star Planetary System: A Time Series Based Analysis of the Planetary Architecture (In Spanish)"*, Licentiate Thesis, Universidad del Valle de Guatemala (2026). Repository Handle.

Research Experience

2026 – Present **Graduate Researcher**, *Department of Physics & Astronomy, University of Delaware, Newark, DE*

Advisor: Prof. Sarah Dodson-Robinson

- Analyzing multi-decadal solar observations to test and calibrate spectral coherence methods for measuring stellar rotation and magnetic activity cycles.
- Developing dual-frequency coherence methods to disentangle stellar activity and magnetic variability from planetary signals in radial velocity datasets.

2025 – 2026 **Undergraduate Visiting Researcher**, *Department of Physics & Astronomy, University of Delaware, Newark, DE*

Advisor: Prof. Sarah Dodson-Robinson; Mentor: Victor Ramirez Delgado

- Evaluated advanced statistical methods to distinguish genuine exoplanetary signals from stellar noise in high-precision radial velocity (RV) data.
- Implemented Gaussian Process (GP) regression and frequency-domain analysis to mitigate stellar variability and improve the reliability of RV detections.
- Analyzed 25 years of multi-instrument RV datasets to determine the planetary architecture of Barnard's Star, utilizing advanced statistical metrics for model validation.

2024 – 2025 **Undergraduate Researcher**, *Department of Physics, Universidad del Valle de Guatemala, Guatemala City*

Advisor: Prof. Eduardo Rubio-Herrera

- Processed and analyzed photometric time-series data from the TESS mission to identify exoplanet candidates.
- Developed an algorithm to detect transits and extract orbital parameters from stellar light curves.

Software

2026 **Flores-Gaitán, O.A.**, *"orbcloud: 3D orbital probability clouds from simulated exoplanet posteriors"*, Zenodo (2026). DOI: 10.5281/zenodo.20935725 GitHub.

- Python library to generate 3D orbital probability clouds from exoplanet MCMC posteriors.

Talks & Posters

Contributed & Invited Talks

2026 **Flores-Gaitán, O.A.**, *"A Dynamically Stable Three-Planet Architecture for Barnard's Star"*, Talk at Exoplanet Group Meeting, Department of Physics & Astronomy, University of Delaware, Newark, DE.

2025 **Flores-Gaitán, O.A.**, Dodson-Robinson, S., Ramirez Delgado, V., Rubio-Herrera, E., *"How many terrestrial planets orbit Barnard's Star?"*, Contributed Talk at APS Mid-Atlantic Section Meeting 2025, Penn State Harrisburg, Middletown, PA. Link.

2025 **Flores-Gaitán, O.A.**, *"Hunting for Exoplanets: Real Planets or Stellar Processes?"*, Invited Talk at Universidad del Valle de Guatemala's Physics Week, Guatemala City.

2025 **Flores-Gaitán, O.A.**, *"Detecting and Characterizing Exoplanets with TESS"*, Invited Talk at Universidad del Valle de Guatemala's Astronomy Club, Guatemala City.

Conference Posters

2025 **Flores-Gaitán, O.A.**, Dodson-Robinson, S., Ramirez Delgado, V., *"Disentangling Planetary and Stellar Signals in Barnard's Star"*, Poster at Physics Science Fair at Universidad del Valle de Guatemala's Physics Week, Guatemala City.

2025 **Flores-Gaitán, O.A.**, Dodson-Robinson, S., Ramirez Delgado, V., *"Disentangling Planetary and Stellar Signals in Barnard's Star"*, Virtual Poster at NASA's Towards the Habitable Worlds Observatory: Visionary Science and Transformational Technology, Johns Hopkins University, Washington, DC.

2025 **Flores-Gaitán, O.A.**, Dodson-Robinson, S., Ramirez Delgado, V., *"Disentangling Planetary and Stellar Signals in Barnard's Star"*, Virtual Poster at Sagan Summer Workshop, NASA Exoplanet Science Institute, California Institute of Technology, Pasadena, CA. Link.

Academic Appointments & Teaching

2026 – Present **Graduate Teaching Assistant**, *Department of Physics & Astronomy, University of Delaware, Newark, DE*

- 4 laboratory sections: PHYS 221 (Introductory Physics Laboratory I).

2024 – 2025 **Undergraduate Teaching Assistant & Laboratory Assistant**, *Physics Department, Universidad del Valle de Guatemala, Guatemala City*

- 7 course sections: Physics I (FF2016), Physics II (FF2017), Physics III (FF2018), and Classical Mechanics I (FF3027).

2022 – Present **Academic Tutor**, *Universidad del Valle de Guatemala, Guatemala City*

- Provided 350+ hours of individual tutoring for high school and university students in Calculus I–III, Differential Equations, and Physics I–III.

Industry & Professional Experience

2026 **Data & Research Analyst**, *Vekst Consulting Group, Guatemala City*

- Led scientific consulting and quantitative modeling projects, applying advanced statistical methods to financial and market data analysis.
- Transformed rigorous research into high-value strategic assets and technical documentation for the consulting sector.
- Designed, developed, and deployed the firm's official website, expanding technical and commercial visibility.

Mentorship

2026 – Present **Isabel Cambronero Jiménez (Universidad de Costa Rica)**, *University of Delaware Undergraduate Research Program, Newark, DE*
Primary Advisor: Prof. Sarah Dodson-Robinson

- Project: Application of the Cepstrum Technique to Exoplanet Radial Velocity Time Series.

Workshops & Summer Schools

2026 **Cohere Labs Machine Learning Summer School**, *Cohere Labs, Remote*

2026 **Code/Astro Workshop**, *University of California, Santa Cruz, Santa Cruz, CA*

2025 **Alpha-Cen Astronomy Summer School**, *Alpha-Cen (Central American and Caribbean Astrophysics), Remote*

2024 **Workshop on Astrophysics of High Energies and Cosmology**, *Universidad Nacional Autónoma de México (UNAM), Guatemala City*

Fellowships, Grants & Awards

2026 **Code/Astro Full Financial Aid Grant**: Awarded full financial support of $1,180 for scientific software development workshop, University of California, Santa Cruz

2026 **Magna Cum Laude**: Graduated with high academic honors in Physics, Universidad del Valle de Guatemala

2025 **Student Travel Award**: Awarded $1,000 travel grant to present research at an international conference, University of Delaware

2025 **Summer Visiting Research Fellowship**: Selected out of 147 international applicants, Department of Physics & Astronomy, University of Delaware

2022, 2023, 2024 **3x Distinguished Student Award (*Estudiante Distinguido*)**: Awarded for top academic standing, Universidad del Valle de Guatemala

2024 **Experimental Physics Record**: Achieved the university's historically most accurate experimental measurement of local gravitational acceleration ($g = 9.7876 \pm 0.0026$ m/s^2), Universidad del Valle de Guatemala

2020 **Top 10 Finalist**: Guatemalan National Mathematics Olympiad

Technical Skills

Programming Languages: Python, LaTeX, R (Intermediate)

Data Analysis & Probabilistic Modeling: Statistical Inference, Gaussian Processes (GPs), Markov Chain Monte Carlo (MCMC), Bayesian Modeling, Time-Series Analysis

Machine Learning: Scikit-learn, XGBoost, Random Forests, Neural Networks, Autoencoders, K-Nearest Neighbors (KNN), Hyperparameter Tuning, Cross-Validation

Astronomy Tools: PyMC, celerite2, exoplanet, AstroPy, Lightkurve, Astroquery, ExoCTK, NWelch

Languages: Spanish (Native), English (C1), German (B2/C1)

Outreach & Leadership

2026 **Invited Panelist**, *CAFE 2026 (Workshop on Astronomical Sciences: Education and Training), Universidad Rafael Landívar, Guatemala City*
- *"Experiences of Internships Abroad for Aspiring Guatemalan Astronomers"*.

2026 **Science Fair Judge**, *Universidad del Valle de Guatemala, Guatemala City*
- Evaluated undergraduate physics and engineering research projects at Universidad del Valle de Guatemala's Science Fair.

2026 **Workshop Developer & Instructor**, *Regional Online Workshop*
- Designed and taught *"How to Apply to Internships in the US"* for undergraduate physics students in Latin America.

2025 **Guest Lecturer & Teaching Assistant**, *Astro Antigua, Antigua Guatemala*
- Collaborated on Extragalactic Astronomy lecture for high school students.

2025 **Lead Organizer**, *Universidad del Valle de Guatemala, Guatemala City*
- Physics Week 2025, a week-long conference and workshop series featuring national and international speakers.

2025 **Workshop Developer & Instructor**, *Universidad del Valle de Guatemala, Guatemala City*
- Designed and taught *"How to Write an Academic CV"* for first-year physics students.

2025 **Active Member**, *Physics Students Association (AEF-UVG)*
- Organizing public lectures and science outreach events.

2024 **Author & Content Creator**, *Universidad del Valle de Guatemala*
- Authored undergraduate Physics laboratory guide and created educational laboratory videos at Universidad del Valle de Guatemala. Link

2021 – 2023 **Founder & Manager**, *Countries Kingdom*
- Educational geography and linguistics outreach project with over **15,000 followers** and **5,000,000+ total views**. Link

Collaborations & Working Groups

- **Extreme Stellar Signals Project (ESSP)**: Collaboration Member through the Sidera Research Group, University of Delaware.
- **Extreme Precision Radial Velocity Research Coordination Network (EPRV RCN)**: Member.

Media

2026 **Actualidad UVG**, *Universidad del Valle de Guatemala, Guatemala City*
- Oscar Flores-Gaitán demonstrates the reach of Physics in the search for exoplanets *(In Spanish)*, Feature Article. Link

2026 **Ciencia UVG Podcast**, *Universidad del Valle de Guatemala*
- A UVG Physicist in Delaware *(In Spanish)*, Podcast Episode. YouTube (Video) / Spotify (Audio)

2024 **Educational Video Tutorial**, *Universidad del Valle de Guatemala, Guatemala City*
- Measuring the gravitational acceleration g with a physical pendulum *(In Spanish)*, YouTube Video. Link